

精算概论第二次作业参考答案

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第一题解答

已知参数: $x = 40, n = 5$, 利率 $i = 0.04$, 贴现因子 $v = 1/1.04 \approx 0.9615$ 。

生存人数: $l_{40} = 200, l_{41} = 195, l_{42} = 188, l_{43} = 180, l_{44} = 170, l_{45} = 160$ 。

死亡人数: $d_{40} = 5, d_{41} = 7, d_{42} = 8, d_{43} = 10, d_{44} = 10$ 。

死亡保额 $B_d = 20000$, 生存保额 $B_s = 10000$ 。

1. 趸缴净保费 (NSP): 死亡给付的期望现值为:

$$\begin{aligned} EPV_{death} &= \sum_{t=1}^5 B_d \cdot v^t \cdot \frac{d_{40+t-1}}{l_{40}} = \frac{20000}{200} \sum_{t=1}^5 v^t d_{39+t} \\ &= 100 \times (5v + 7v^2 + 8v^3 + 10v^4 + 10v^5) \\ &= 100 \times (4.8077 + 6.4719 + 7.1120 + 8.5480 + 8.2193) \\ &= 100 \times 35.1589 = 3515.89 \text{ (元)} \end{aligned}$$

生存给付的期望现值为:

$$\begin{aligned} EPV_{survival} &= B_s \cdot v^5 \cdot \frac{l_{45}}{l_{40}} = 10000 \times (0.961538)^5 \times \frac{160}{200} \\ &= 10000 \times 0.821927 \times 0.8 = 6575.42 \text{ (元)} \end{aligned}$$

趸缴净保费 $A = 3515.89 + 6575.42 = 10091.31$ (元)。

2. 5 年期年缴净保费 (P_5): 5 年期期初生存年金现值:

$$\begin{aligned} \ddot{a}_{40:\overline{5}|} &= 1 + v \frac{l_{41}}{l_{40}} + v^2 \frac{l_{42}}{l_{40}} + v^3 \frac{l_{43}}{l_{40}} + v^4 \frac{l_{44}}{l_{40}} \\ &= 1 + 0.9615 \times \frac{195}{200} + 0.9246 \times \frac{188}{200} + 0.8890 \times \frac{180}{200} + 0.8548 \times \frac{170}{200} \\ &= 1 + 0.9375 + 0.8691 + 0.8001 + 0.7266 = 4.3333 \end{aligned}$$

$$P_5 = \frac{A}{\ddot{a}_{40:\overline{5}|}} = \frac{10091.31}{4.3333} \approx 2328.78 \text{ (元)}。$$

3. 3 年期年缴净保费 (P_3): 3 年期期初生存年金现值:

$$\ddot{a}_{40:\overline{3}|} = 1 + v \frac{l_{41}}{l_{40}} + v^2 \frac{l_{42}}{l_{40}} = 1 + 0.9375 + 0.8691 = 2.8066$$

$$P_3 = \frac{A}{\ddot{a}_{40:\overline{3}|}} = \frac{10091.31}{2.8066} \approx 3595.56 \text{ (元)}。$$

4. 3 年期年缴毛保费 (G): 设费用比例依次为 $\alpha_0 = 0.35, \alpha_1 = 0.15, \alpha_2 = 0.10$ 。依据收支相等

原则：

$$G \cdot \ddot{a}_{40:\overline{3}|} = A + G (0.35 + 0.15vp_{40} + 0.10v^2{}_2p_{40})$$

$$G \cdot [1 - 0.35 + (1 - 0.15)vp_{40} + (1 - 0.10)v^2{}_2p_{40}] = A$$

$$G \cdot [0.65 + 0.85 \times 0.9375 + 0.90 \times 0.8691] = 10091.31$$

$$G \cdot (0.65 + 0.7969 + 0.7822) = 10091.31$$

$$G \cdot 2.2291 = 10091.31 \implies G \approx \mathbf{4527.08 \text{ (元)}}$$

第二题解答

已知参数: $x = 32, n = 3, i = 0.035, v = 1/1.035 \approx 0.96618, B = 50000$ 。

根据生命表: $q_{32} = 0.000903 \implies p_{32} = 0.999097$

$q_{33} = 0.000966 \implies p_{33} = 0.999034$

$q_{34} = 0.001035 \implies p_{34} = 0.998965$

可计算生存概率: ${}_1p_{32} = 0.999097, {}_2p_{32} = p_{32}p_{33} \approx 0.998132, {}_3p_{32} = {}_2p_{32}p_{34} \approx 0.997099$ 。

1. 趸缴净保费 (NSP): 死亡给付现值:

$$\begin{aligned} EPV_{death} &= 50000 \times (vq_{32} + v^2p_{32}q_{33} + v^3{}_2p_{32}q_{34}) \\ &= 50000 \times [(0.96618)(0.000903) + (0.93351)(0.999097)(0.000966) + (0.90194)(0.998132)(0.001035)] \\ &= 50000 \times (0.000872 + 0.000901 + 0.000932) = 50000 \times 0.002705 = 135.25 \text{ (元)} \end{aligned}$$

生存给付现值:

$$EPV_{survival} = 50000 \times v^3 \times {}_3p_{32} = 50000 \times 0.90194 \times 0.997099 = 44966.18 \text{ (元)}$$

趸缴净保费 $A = 135.25 + 44966.18 = 45101.43 \text{ (元)}$ 。

2. 3 年期年缴净保费:

$$\begin{aligned} \ddot{a}_{32:\overline{3}|} &= 1 + vp_{32} + v^2{}_2p_{32} \\ &= 1 + (0.96618)(0.999097) + (0.93351)(0.998132) \\ &= 1 + 0.96531 + 0.93177 = 2.89708 \end{aligned}$$

年缴净保费 $P = \frac{45101.43}{2.89708} \approx 15567.89 \text{ (元)}$ 。

第三题解答

已知参数: $T \sim U(0, 80)$, $f_T(t) = 1/80$, $\delta = 0.05$, $Z = e^{-\delta T}$ 。

1. 计算 $\bar{A}_x$:

$$\bar{A}_x = E[Z] = \int_0^{80} e^{-0.05t} \cdot \frac{1}{80} dt = \frac{1}{80} \left[\frac{e^{-0.05t}}{-0.05} \right]_0^{80} = \frac{1}{-4} (e^{-4} - 1) = \frac{1 - e^{-4}}{4} \approx \mathbf{0.2454}$$

2. 计算 $\xi_{0.95}$:

$$P(Z \leq \xi_{0.95}) = 0.95 \implies P(e^{-0.05T} \leq \xi_{0.95}) = 0.95$$

$$P(-0.05T \leq \ln \xi_{0.95}) = 0.95 \implies P\left(T \geq -\frac{\ln \xi_{0.95}}{0.05}\right) = 0.95$$

因为 $T \sim U(0, 80)$, 要使得大于某值的概率为 0.95, 该值必定为 $80 \times (1 - 0.95) = 4$ 。

$$-\frac{\ln \xi_{0.95}}{0.05} = 4 \implies \ln \xi_{0.95} = -0.2 \implies \xi_{0.95} = e^{-0.2} \approx \mathbf{0.8187}$$

第四题解答

已知参数: $f_T(t) = 0.05e^{-0.05t}$, $\mu = 0.05$, $\delta = 0.04$ 。

1. 趸缴净保费 $\bar{A}_x$:

$$\bar{A}_x = E[Z] = \int_0^\infty e^{-\delta t} \mu e^{-\mu t} dt = \frac{\mu}{\mu + \delta} = \frac{0.05}{0.05 + 0.04} = \frac{5}{9} \approx \mathbf{0.5556}$$

2. 计算 $\xi_{0.99}$:

$$P(Z \leq \xi_{0.99}) = 0.99 \implies P\left(T \geq -\frac{\ln \xi_{0.99}}{0.04}\right) = 0.99$$

由指数分布特性, $P(T \geq t_0) = e^{-0.05t_0} = 0.99 \implies t_0 = -20 \ln(0.99) \approx 0.2010$ 。

$$-\frac{\ln \xi_{0.99}}{0.04} = -20 \ln(0.99) \implies \ln \xi_{0.99} = 0.8 \ln(0.99) \implies \xi_{0.99} = (0.99)^{0.8} \approx \mathbf{0.9920}$$

3. 中心极限定理应用: 首先求单一保单的方差:

$$E[Z^2] = \int_0^\infty e^{-2\delta t} \mu e^{-\mu t} dt = \frac{\mu}{\mu + 2\delta} = \frac{0.05}{0.05 + 0.08} = \frac{5}{13} \approx 0.3846$$

$$Var(Z) = E[Z^2] - (E[Z])^2 = \frac{5}{13} - \left(\frac{5}{9}\right)^2 = \frac{405 - 325}{1053} = \frac{80}{1053} \approx 0.0760$$

对于 $N = 2000$ 份保单的赔付总额 $S = \sum Z_i$:

$$E[S] = 2000 \times \frac{5}{9} \approx 1111.11$$

$$Var(S) = 2000 \times \frac{80}{1053} \approx 151.95 \implies \sigma_S = \sqrt{151.95} \approx 12.33$$

设初始基金为 F , 满足 $P(S \leq F) = 0.99$, 查标准正态分布表得 $z_{0.99} \approx 2.326$ 。

$$F = E[S] + 2.326 \cdot \sigma_S = 1111.11 + 2.326 \times 12.33 = 1111.11 + 28.68 = \mathbf{1139.79 \text{ (元)}}$$

第五题解答

1. 单位终身寿险 (Z_1):

$$E(Z_1) = \int_0^{\infty} e^{-\delta t} \mu e^{-\mu t} dt = \frac{\mu}{\mu + \delta}$$

$$E(Z_1^2) = \int_0^{\infty} e^{-2\delta t} \mu e^{-\mu t} dt = \frac{\mu}{\mu + 2\delta}$$

$$\text{Var}(Z_1) = E(Z_1^2) - [E(Z_1)]^2 = \frac{\mu}{\mu + 2\delta} - \left(\frac{\mu}{\mu + \delta} \right)^2$$

2. 单位 n 年期两全保险 (Z_2): 现值随机变量 $Z_2 = \begin{cases} e^{-\delta T}, & 0 \leq T \leq n \\ e^{-\delta n}, & T > n \end{cases}$

$$E(Z_2) = \int_0^n e^{-\delta t} \mu e^{-\mu t} dt + e^{-\delta n} \int_n^{\infty} \mu e^{-\mu t} dt$$

$$= \frac{\mu}{\mu + \delta} \left(1 - e^{-(\mu + \delta)n} \right) + e^{-\delta n} \cdot e^{-\mu n}$$

$$= \frac{\mu}{\mu + \delta} - \frac{\mu}{\mu + \delta} e^{-(\mu + \delta)n} + e^{-(\mu + \delta)n}$$

$$E(Z_2^2) = \int_0^n e^{-2\delta t} \mu e^{-\mu t} dt + e^{-2\delta n} P(T > n)$$

$$= \frac{\mu}{\mu + 2\delta} \left(1 - e^{-(\mu + 2\delta)n} \right) + e^{-(\mu + 2\delta)n}$$

$$\text{Var}(Z_2) = E(Z_2^2) - [E(Z_2)]^2$$

$$= \frac{\mu}{\mu + 2\delta} \left(1 - e^{-(\mu + 2\delta)n} \right) + e^{-(\mu + 2\delta)n} - \left[\frac{\mu}{\mu + \delta} \left(1 - e^{-(\mu + \delta)n} \right) + e^{-(\mu + \delta)n} \right]^2$$

第六题解答

已知参数: $x = 39$, 递延期 $n = 45 - 39 = 6$ 年, $\mu = 0.008, \delta = 0.04$, 保额 $B = 2,000,000$ 。
这是一个 n 年期纯生存保险 (Pure Endowment)。其期望现值为:

$$\begin{aligned} EPV &= B \cdot e^{-\delta n} \cdot {}_n p_x \\ &= B \cdot e^{-\delta n} \cdot e^{-\mu n} \\ &= B \cdot e^{-(\mu+\delta)n} \end{aligned}$$

代入具体数值:

$$\begin{aligned} EPV &= 2000000 \cdot e^{-(0.008+0.04) \times 6} \\ &= 2000000 \cdot e^{-0.048 \times 6} = 2000000 \cdot e^{-0.288} \end{aligned}$$

已知 $e^{-0.288} \approx 0.74976$, 计算得:

$$EPV = 2000000 \times 0.74976 = \mathbf{1499520 \text{ (元)}}$$